Exponents, Squares and Square Roots

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KEY WORDS exponent *(taupū)* · base *(pūtake)* · square number *(tau pūrua)* · square root *(pūtakerua)*

A Exponent notation ●○○

⚠ **Watch out:** The exponent tells you how many times the base is multiplied by itself. 2^3 means $2 \times 2 \times 2 = 8$. It does not mean 2×3.

1. Write each multiplication using an exponent.

a) $5 \times 5 \times 5$ = _____ **b)** $2 \times 2 \times 2 \times 2$ = _____

c) 7×7 = _____ **d)** $10 \times 10 \times 10 \times 10 \times 10$ = _____

e) $3 \times 3 \times 3 \times 3 \times 3 \times 3$ = _____ **f)** $9 \times 9 \times 9$ = _____

2. Work out each power. Write the multiplication first.

a) 2^3 = _____ **b)** 3^2 = _____

c) 10^3 = _____ **d)** 2^5 = _____

e) 5^3 = _____ **f)** 1^6 = _____

g) 3^4 = _____ **h)** 4^3 = _____

B Square numbers ●○○

💡 **Remember:** To square a number, multiply it by itself: $7^2 = 7 \times 7 = 49$. The answers are called square numbers.

3. Work out.

a) 3^2 = _____ **b)** 6^2 = _____ **c)** 1^2 = _____ **d)** 9^2 = _____

e) 4^2 = _____ **f)** 11^2 = _____ **g)** 7^2 = _____ **h)** 12^2 = _____

i) 5^2 = _____ **j)** 10^2 = _____ **k)** 8^2 = _____ **l)** 2^2 = _____

m) $3^2 + 4^2$ = _____ **n)** $10^2 - 8^2$ = _____ **o)** 2×5^2 = _____ **p)** $6^2 + 8^2$ = _____

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C Square roots ●●○

Remember: The square root undoes squaring: $\sqrt{49} = 7$ because $7^2 = 49$.

4. Work out.

a) $\sqrt{25}$ = _____ **b)** $\sqrt{81}$ = _____ **c)** $\sqrt{4}$ = _____ **d)** $\sqrt{100}$ = _____

e) $\sqrt{36}$ = _____ **f)** $\sqrt{121}$ = _____ **g)** $\sqrt{1}$ = _____ **h)** $\sqrt{64}$ = _____

i) $\sqrt{144}$ = _____ **j)** $\sqrt{9}$ = _____ **k)** $\sqrt{49}$ = _____ **l)** $\sqrt{16}$ = _____

m) $\sqrt{9} + \sqrt{16}$ = ____ **n)** $\sqrt{100} - \sqrt{36}$ = ____ **o)** $2 \times \sqrt{25}$ = _____ **p)** $\sqrt{9 + 16}$ = _____

D Working with powers ●●●

Remember: Work out each power first, then add, subtract or multiply.

5. Work out.

a) $2^3 + 3^2$ = _____ **b)** $10^2 - 5^2$ = _____ **c)** 4^3 = _____ **d)** $2^4 \times 3$ = _____

e) $5^2 \times 2^2$ = _____ **f)** $3^3 - 2^3$ = _____ **g)** $10^3 + 10^2$ = _____ **h)** $6^2 \div 4$ = _____

i) $2^5 - 5^2$ = _____ **j)** $1^7 + 7^1$ = _____ **k)** $\sqrt{81} + 2^3$ = _____ **l)** $\sqrt{100} \times 3^2$ = _____

E Using squares and square roots ●●●

6. Find the area of a square with sides of:

a) 6 cm **b)** 11 cm **c)** 20 cm **d)** 15 cm

7. Find the length of the sides of a square with an area of:

a) 49 cm² **b)** 100 cm² **c)** 36 cm² **d)** 400 cm²

8. A square has sides of 9 cm. Find its area.

9. A square has an area of 121 cm². How long is each side?

10. Which two square numbers is each number between?

a) 20 **b)** 50 **c)** 90 **d)** 130

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11. Add these odd numbers. What do you notice about the answers?

a) $1 + 3$ **b)** $1 + 3 + 5$ **c)** $1 + 3 + 5 + 7$

12. Use what you noticed to write down the sum of the first 10 odd numbers.

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Exponents, Squares and Square Roots

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS exponent *(taupū)* · base *(pūtake)* · square number *(tau pūrua)* · square root *(pūtakerua)*

A Exponent notation ●○○

⚠ **Watch out:** The exponent tells you how many times the base is multiplied by itself. 2^3 means $2 \times 2 \times 2 = 8$. It does not mean 2×3.

1. Write each multiplication using an exponent.

a) $4 \times 4 \times 4$ = _____ **b)** $3 \times 3 \times 3 \times 3$ = _____

c) 8×8 = _____ **d)** $10 \times 10 \times 10 \times 10$ = _____

e) $2 \times 2 \times 2 \times 2 \times 2 \times 2$ = _____ **f)** $6 \times 6 \times 6$ = _____

2. Work out each power. Write the multiplication first.

a) 2^4 = _____ **b)** 4^2 = _____

c) 10^4 = _____ **d)** 3^3 = _____

e) 5^2 = _____ **f)** 1^8 = _____

g) 2^6 = _____ **h)** 6^3 = _____

B Square numbers ●○○

💡 **Remember:** To square a number, multiply it by itself: $7^2 = 7 \times 7 = 49$. The answers are called square numbers.

3. Work out.

a) 4^2 = _____ **b)** 7^2 = _____ **c)** 1^2 = _____ **d)** 8^2 = _____

e) 3^2 = _____ **f)** 12^2 = _____ **g)** 6^2 = _____ **h)** 11^2 = _____

i) 5^2 = _____ **j)** 10^2 = _____ **k)** 9^2 = _____ **l)** 2^2 = _____

m) $6^2 + 2^2$ = _____ **n)** $10^2 - 6^2$ = _____ **o)** 2×4^2 = _____ **p)** $5^2 + 12^2$ = _____

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C Square roots ●●○

Remember: The square root undoes squaring: $\sqrt{49} = 7$ because $7^2 = 49$.

4. Work out.

a) $\sqrt{16}$ = _____ **b)** $\sqrt{49}$ = _____ **c)** $\sqrt{9}$ = _____ **d)** $\sqrt{100}$ = _____

e) $\sqrt{64}$ = _____ **f)** $\sqrt{144}$ = _____ **g)** $\sqrt{1}$ = _____ **h)** $\sqrt{81}$ = _____

i) $\sqrt{121}$ = _____ **j)** $\sqrt{4}$ = _____ **k)** $\sqrt{36}$ = _____ **l)** $\sqrt{25}$ = _____

m) $\sqrt{4} + \sqrt{49}$ = _____ **n)** $\sqrt{144} - \sqrt{81}$ = _____ **o)** $3 \times \sqrt{16}$ = _____ **p)** $\sqrt{36 + 64}$ = _____

D Working with powers ●●●

Remember: Work out each power first, then add, subtract or multiply.

5. Work out.

a) $3^2 + 2^4$ = _____ **b)** $10^2 - 6^2$ = _____ **c)** 5^3 = _____ **d)** $2^3 \times 5$ = _____

e) $4^2 \times 2^2$ = _____ **f)** $3^3 - 4^2$ = _____ **g)** $10^3 + 10^1$ = _____ **h)** $8^2 \div 4$ = _____

i) $2^6 - 6^2$ = _____ **j)** $1^9 + 9^1$ = _____ **k)** $\sqrt{64} + 3^2$ = _____ **l)** $\sqrt{49} \times 2^3$ = _____

E Using squares and square roots ●●●

6. Find the area of a square with sides of:

a) 7 cm **b)** 12 cm **c)** 30 cm **d)** 14 cm

7. Find the length of the sides of a square with an area of:

a) 64 cm² **b)** 81 cm² **c)** 25 cm² **d)** 900 cm²

8. A square has sides of 8 cm. Find its area.

9. A square has an area of 144 cm². How long is each side?

10. Which two square numbers is each number between?

a) 30 **b)** 40 **c)** 70 **d)** 110

★ Extension

11. Add these odd numbers. What do you notice about the answers?

a) $1 + 3 + 5$ **b)** $1 + 3 + 5 + 7$ **c)** $1 + 3 + 5 + 7 + 9$

12. Use what you noticed to write down the sum of the first 8 odd numbers.

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